

Track B vs Track C Comparison Report

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Checkpoint: Weighted-class only (0.3708)

Scoring: Composite score = mAP + N_top5_mAP + Nv_top5_mAP + N_delta_top5_mAP + All_top5_mAP

Executive Summary

Best Runs Comparison (by Composite Score)

Metric	Track B Best	Track C Best	Winner	Δ
Composite Score	0.7507	0.7121	Track B	-0.0386
mAP	37.34%	37.55%	Track C	+0.21%
accuracy	64.23%	65.03%	Track C	+0.80%
All_top5_mAP	4.81%	4.10%	Track B	-0.70%
N_top5_mAP	15.28%	13.65%	Track B	-1.63%
Nv_top5_mAP	4.81%	4.36%	Track B	-0.45%
N_delta_top5_mAP	12.83%	11.55%	Track B	-1.29%
All_top5_acc	3.00%	2.58%	Track B	-0.43%
N_top5_acc	18.67%	18.24%	Track B	-0.43%
ttc_mae_seconds	0.1996	0.1989	Track C	-0.0008

Track C Best Latency: 23.6ms (43% faster than baseline)

Best Track B Run

run	checkpoint	score	mAP	accuracy	N_top5_mAP	Nv_top5_mAP	N_delta_top5_mAP
metrics_val_20251226_231700_summary.json	trackB_best_mAP_0.3708_20251225_224220.pt	0.7507	37.34%	64.23%	15.28%	4.81%	12.83%

Best Track C Runs by Category

Category	Config	score	mAP	All_top5	N_top5	Nv_top5	N_delta	Latency	Speedup
Best Composite Score	Fr=8(uni)	0.7121	37.55%	4.10%	13.65%	4.36%	11.55%	23.6499	43%
Best All_top5_mAP	Fr=8(uni)	0.7121	37.55%	4.10%	13.65%	4.36%	11.55%	23.6499	43%
Best N_top5_mAP	Fr=4(uni) + Tok=0.3	0.7068	33.46%	2.19%	16.70%	2.19%	16.13%	19.8198	52%
Best Nv_top5_mAP	Fr=8(uni)	0.7121	37.55%	4.10%	13.65%	4.36%	11.55%	23.6499	43%
Best mAP	Fr=8(mot)	0.7008	37.74%	2.66%	14.36%	2.66%	12.65%	23.8724	42%

Pareto-Optimal Configurations (Score vs Latency)

These configurations are NOT dominated by any other run:

Rank	Config	score	All_top5	N_top5	Nv_top5	mAP	Latency	Speedup
1	Fr=8(uni)	0.7121	4.10%	13.65%	4.36%	37.55%	23.1193	44%
2	Fr=4(uni) + Tok=0.3	0.7068	2.19%	16.70%	2.19%	33.46%	19.8198	52%

All Track B Runs (Weighted Checkpoint)

run	checkpoint	score	mAP	accuracy	N_top5_mAP	Nv_top5_mAP	N_delta_top5_mAP
metrics_val_20251226_231700_summary.json	trackB_best_mAP_0.3708_20251225_224220.pt	0.7507	37.34%	64.23%	15.28%	4.81%	12.83%
metrics_val_20251226_013919_summary.json	trackB_best_mAP_0.3708_20251225_224220.pt	0.7446	37.34%	64.23%	15.01%	4.73%	12.65%
metrics_val_20251227_001540_summary.json	trackB_best_mAP_0.3708_20251225_224220.pt	0.6872	37.34%	64.23%	14.15%	2.88%	11.41%

All Track C Runs (Weighted Checkpoint)

timestamp	Config	score	mAP	accuracy	All_top5	N_top5	Nv_top5	N_delta	Latency	TTC MAE
225122	Fr=8(uni)	0.7121	37.55%	65.03%	4.10%	13.65%	4.36%	11.55%	23.6499	0.1989
232658	Fr=8(uni)	0.7121	37.55%	65.03%	4.10%	13.65%	4.36%	11.55%	23.1193	0.1989
214204	Fr=4(uni) + Tok=0.3	0.7068	33.46%	67.68%	2.19%	16.70%	2.19%	16.13%	19.8198	0.1964
234937	Fr=4(uni) + Tok=0.3	0.7068	33.46%	67.68%	2.19%	16.70%	2.19%	16.13%	20.0264	0.1964
141723	Baseline	0.7010	37.61%	64.76%	3.44%	13.80%	3.35%	11.89%	40.7248	0.1989
161030	Baseline	0.7010	37.61%	64.76%	3.44%	13.80%	3.35%	11.89%	41.8309	0.1989
200931	Fr=8(mot)	0.7008	37.74%	64.90%	2.66%	14.36%	2.66%	12.65%	23.8724	0.2001
151551	Baseline	0.7006	37.61%	64.76%	3.44%	13.79%	3.34%	11.88%	41.2440	0.1989
153959	Baseline	0.7006	37.61%	64.76%	3.44%	13.79%	3.34%	11.88%	41.7022	0.1989
163017	RGTP=0.1	0.6988	37.35%	64.37%	3.45%	13.81%	3.36%	11.92%	25.2287	0.1992
193527	RGTP=0.1	0.6988	37.35%	64.37%	3.45%	13.81%	3.36%	11.92%	23.2121	0.1992
165938	RGTP=0.1	0.6985	37.35%	64.37%	3.44%	13.80%	3.35%	11.91%	25.5391	0.1992
211854	Tok=0.3	0.6760	33.10%	67.82%	2.40%	15.46%	2.40%	14.24%	22.4142	0.1978
172547	RGTP=0.3	0.6732	35.45%	54.41%	3.48%	13.30%	3.33%	11.76%	24.5602	0.1998
210351	Tok=0.1	0.6260	32.73%	67.29%	2.05%	13.66%	2.05%	12.11%	22.9234	0.1972
202451	Fr=8(mot) + Tok=0.1	0.5754	33.09%	67.35%	1.59%	11.38%	1.59%	9.89%	22.9530	0.1981

Trade-off Analysis

Key Trade-off Insights

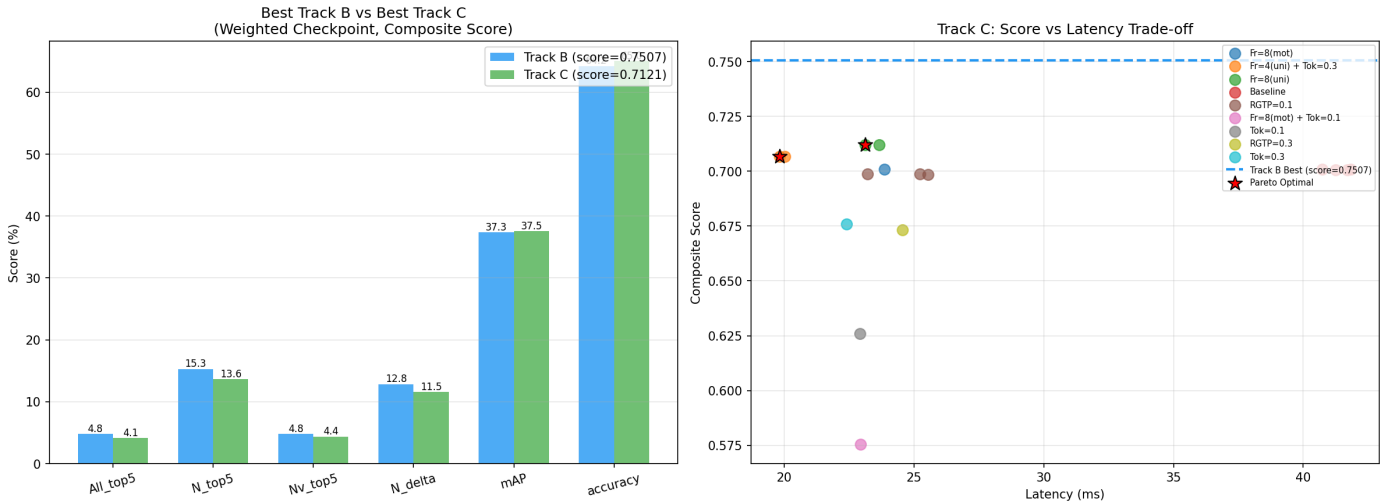
1. **Best All_top5_mAP:** 4.10% (Fr=8(uni))
 - vs Track B: -0.70%
 - Latency: 23.6ms (43% faster)
2. **Best N_top5_mAP:** 16.70% (Fr=4(uni) + Tok=0.3)
 - vs Track B: +1.43% 📈
3. **Maximum Speedup:** 52% (Fr=4(uni) + Tok=0.3)
 - Latency: 19.8ms (vs 41.4ms baseline)
 - All_top5: 2.19%
 - Score: 0.7068

Recommendations

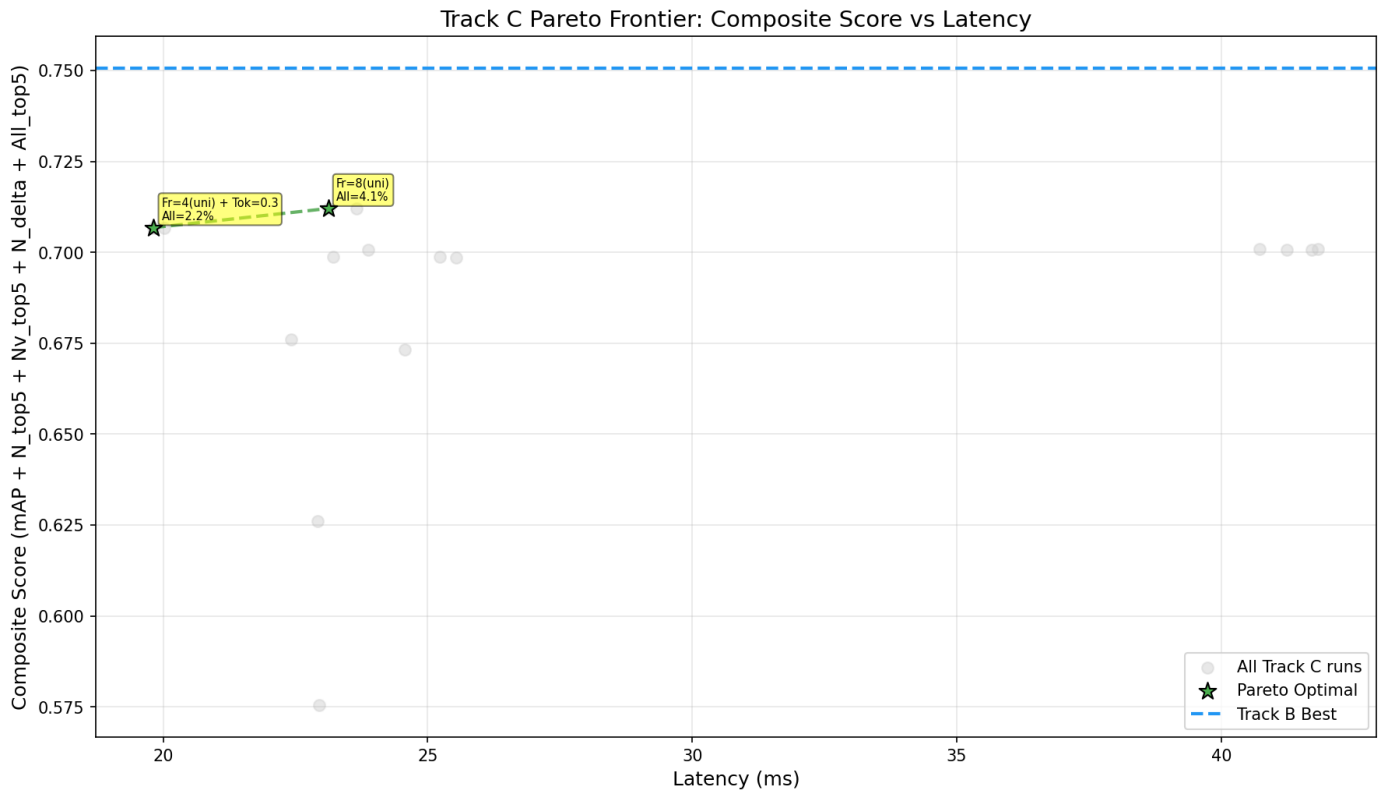
Use Case	Recommended Config	Score	All_top5	Latency	Speedup
Max Score	Fr=8(uni)	0.7121	4.10%	23.1ms	44%
Best Trade-off	Fr=8(uni)	0.7121	4.10%	23.1ms	44%
Max Efficiency	Fr=4(uni) + Tok=0.3	0.7068	2.19%	19.8ms	52%

Plots

Track B vs Track C Comparison



Pareto Frontier (Score vs Latency)



Individual Metrics Trade-off

